

Society V1

Anatomy of the 42

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The 42 conceptual variables that compose Society V1 represent abstractions of the fundamental forces operating within society. These concepts (e.g., DIVERSITY, POVERTY, MENTAL HEALTH, etc.) are mathematically defined by an influence matrix, a collection of output edges with weighted polarity expressed as normalized values between -1 and 1. Each variable is materially tethered to a physical slider on the Society V1 controller board, allowing users to interact directly with the model.

Changing the value of a variable by moving its corresponding slider adjusts the directional intensity of the influence it exerts on others connected to it. No change occurs in isolation; rather, it incites conditional responses along associated pathways, dynamically reshaping the collective state of the system.

To illustrate: If you change the value of LAW ENFORCEMENT, depending on the current vector state of the system, CRIME will change its state by an amount determined by the weighted edge connecting the two variables. When CRIME settles into its new position, its own network of influences will trigger, affecting downstream variables like DEATH RATE and PUBLIC TRUST.

The mechanism may appear straightforward, but it naturally raises important questions. How are the weights assigned? What justifications or rationale support the exact values of these weights? Are they speculative? Is it arbitrary? How can a specific value be deduced from empirical research when the findings themselves often present convoluted observations or abstract outcomes inferred by researchers?

This document outlines how the values for these concepts are informed by and inferred from aggregated research findings, comparative data, and disciplined observational reasoning.

For readers interested in the broader overview of the system, see *Society V1: Whitepaper*. For in-depth technical implementation, including the influence engine and mathematical foundations, see *Computational & Mathematical Architecture (CMA)*.

It should be underscored that these values, along with the mathematics directing them, are ultimately transcribed by a single mortal mind¹ attempting, through large bodies of research and literature, to painstakingly quantify abstractions far more complex than any model can fully capture. The act of bounding these broad abstractions into discrete values inevitably reduces their individual richness; however, at scale, when enough are structured together, this reduction becomes the very mechanism that makes modeling society possible.

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|----------------------|------------------------|---------------------|----------------------|
| 1. Addiction | 12. Education Cost | 23. Homelessness | 34. Poverty |
| 2. Agriculture | 13. Education Quality | 24. Immigration | 35. Public Trust |
| 3. Authoritarianism | 14. Employment Rate | 25. Infrastructure | 36. Rule of Law |
| 4. Birthrate | 15. Energy Production | 26. Innovation | 37. Spirituality/God |
| 5. Corruption | 16. Equality | 27. Law Enforcement | 38. Sickness/Disease |
| 6. Cost of Living | 17. Family Orientation | 28. Liberty | 39. Taxes |
| 7. Crime | 18. Freedom of Speech | 29. Market Freedom | 40. Technology |
| 8. Death Rate | 19. Gun Rights | 30. Mental Health | 41. Transportation |
| 9. Discrimination | 20. Healthcare Access | 31. Military | 42. Welfare Support |
| 10. Diversity | 21. Healthcare Cost | 32. Minimum Wage | |
| 11. Education Access | 22. Healthcare Quality | 33. Pollution | |

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¹Currently developed by a single human modeler, though additional minds would likely improve both the model and the modeler.

► Interpretation Framework

Before jumping down the rabbit hole, I feel it would be wise to provide the reader with context for how each variable is interpreted and structured. To avoid confusion, all variable descriptions in this document will follow a consistent formatting framework. Each section will include a detailed description, methodological scope, and acknowledgment of any known challenges or disputes within the academic literature.

Ontology

Society does not consist of cleanly separable components with absolute borders or dividing lines. In reality, many social phenomena overlap, interact, repel, and co-evolve in extremely complex ways. Because of this, clear conceptual distinctions must be intentionally drawn in order to model them.

For the purposes of Society V1, each variable is assigned abstract boundaries. In the sections that follow, these abstractions are outlined and systematized to minimize conceptual overlap while preserving the ability to model emergent system behavior with reasonable fidelity.

Inclusionary & Exclusionary Scopes

Inclusionary and exclusionary lists define the conceptual perimeter of each variable. These boundaries clarify what is intentionally considered and what is not included within the boundaries of each conceptual variable, ensuring shared interpretation and internal consistency.

This document aims to answer practical questions such as:

- › What does each variable (or their numerical values) actually mean?
 - › What am I changing when I move a variable's value up or down?
 - › How are these concepts measured in the real world?
 - › How are specific influence values determined when research findings vary across studies?
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Research Metrics & Methodology²

All sections will contain the relevant quantitative metrics, secondary indicators, and research approaches corresponding to the verifiable data used to ground each variable's influence weight to the real world.

Model parameters are derived through a structured synthesis of:

- › Established empirical research
- › Cross-national and national statistical datasets
- › Peer-reviewed literature
- › AI tools³
- › Comparative historical analysis

Path-Dependent Influence Weights

Variables are defined broadly, but their output pathways (influence edges) are formulated contextually. Each variable influences others differently, depending on the pathway through which the interaction occurs. In other words, every variable casts different shadows depending on where it lands. Accordingly, each influence edge weight is inferred from the empirical metrics most directly associated with the connection being modeled.

For example, ADDICTION is not simply a singular influence weight applied uniformly across different receptor variables (such as CRIME or LIBERTY). Instead, it functions as a matrix of path-specific influence weights that exert discretely different intensities across its downstream relationships. ADDICTION influence on DEATH RATE is evaluated through overdose and attributable mortality, while its influence on MENTAL HEALTH is interpreted through the prevalence of dual-diagnosis, surveys, and hospitalization rates.

In this way, the model maintains empirical grounding while avoiding the false assumption that a complex societal variable can be represented by a single uniform measurement.

²To find Society V1's citations, datasets, literature, and research sources see: RESEARCH REPOSITORY

³Including platforms such as Consensus AI for literature aggregation and comparative analysis.

1 Addiction

ADDICTION is a variable that encompasses the severity and pervasiveness of two primary dependency domains: Substance Use Disorders and Behavioral Disorders.

Dependency to substances (e.g., alcohol, opioids, stimulants, tobacco) is typically characterized by physiological adaptation; tolerance and withdrawal. In contrast, behavioral addictions (e.g., gambling, sex, pornography, digital dependency) are pattern-reinforced compulsions driven by endogenous reward systems. They lack direct pharmacological dependence and their withdrawal tends to be psychological rather than physiological.

Although the mechanisms between chemical and behavioral dependencies can differ, both produce an equally persistent psychological grip, and both are strongly defined by the phenomena of craving.

Within the domains of substance use and behavioral disorders, there are two separate dimensions: *Physiological Dependency* and *Psychological Compulsion*.

This variable is modeled by placing emphasis on the relevant mechanisms of addiction that propagate into broader societal effects, at the population level, rather than individual experiences at the psychological level.

» ADDICTIONincludes:

- › Pervasiveness of active addiction
- › Severity and chronicity
- › Recovery and relapse dynamics
- › Economic, judicial, and social burdens
- › Social and institutional degradation

» ADDICTIONexcludes:

- › General mental illnesses
- › Occasional recreational use
- › Addiction-related mortality
- › Non-compulsive binge behavior
- › Criminality

When the value of ADDICTIONincreases, it represents a greater prevalence and severity of dependency within the population; conversely, when it decreases, those burdens recede.

In Society V1, ADDICTIONfunctions as both a public health burden and a social destabilizer, influencing productivity, MENTAL HEALTH, family stability, CRIME, and broader societal resilience. ADDICTIONis supported by a robust empirical foundation, allowing it to be modeled with a relatively high degree of confidence.

Research Metrics & Methodology

Primary	Secondary
Substance Use Disorder (SUD) prevalence	Self-reported dependency surveys
Overdose rates	Drug-related economic burden estimates
Treatment enrollment rates	Workplace impairment statistics
Relapse rates	Dual Diagnosis Prevalence (psychiatric comorbidity)
Hospitalization Statistics	

Data Challenges & Academic Disputes

- › Under-reporting
- › Inconsistencies in cross-national reporting
- › Measurement gaps
- › Cultural normalization differences
- › Social stigma; dampening treatment and diagnosis visibility
- › Misdiagnosis; also including Over- and Under-diagnosis
- › Criminalization vs public health framing
- › Direction of Socioeconomic causality: (Poverty → Addiction) vs. (Addiction → Poverty)
- › Measurement inconsistency in clinical and survey definitions

The empirical data on addiction could be accurately illustrated with a slice of Swiss cheese. Let's point out some of the holes: For one, surveys cannot record what goes unreported. Bias from Institutional incentives cannot be untangled from those made from financial pressures. Misdiagnoses, whether from inept or complacent professionals, misshape the statistics we are supposed to trust in a significant way. Add to that the inconsistent standards across country-to-country data sharing, and the data we are left with is a distorted picture, a hole, in the very reality it attempts to represent. Therefore, empirical data on addiction = Swiss cheese.

Unfortunately, because of how it manifests, there will always be disputes over whether addiction is more of a public health, criminal justice, or even a genetic issue; which shifts the goalposts for how each case is understood. And it seems that the more standardized a treatment is, the less it works long-term. Our lack of understanding is evident in the growing number of addiction diagnoses, despite the exponential growth of treatment programs and their budgets.

Active Output Edges: ADDICTION

Output Edges	Weight	Mechanism Summary
Agriculture	-0.08	Small reduction; erodes agricultural productivity and increases injury risk
Birthrate	-0.16	Moderate reduction; diminishing returns become offset by increased unplanned pregnancies
Cost of Living	+0.15	Moderate increase; healthcare costs, productivity loss, and financial irresponsibility
Crime	+0.30	Strong increase; desperation, impaired judgment, and social destabilization
Death Rate	+0.31	Heavy increases mortality rate; overdoses, disease progression, accidents, suicides, and violence
Education Access	-0.07	Small but noticeable decrease; reduced enrollment, attendance, and increased disciplinary exclusion
Education Quality	-0.17	Moderate decrease; impairs concentration, absenteeism, disruption, lower engagement
Employment	-0.23	Substantial reduction; degrades reliability, productivity, and hiring prospects
Equality	-0.11	Slight erosion; concentrates harm in disadvantaged groups and suppresses mobility
Family Orientation	-0.21	Substantial reduction; erodes families, increases conflict, and dilutes inter-generational values
Gun Rights	-0.06	Small decrease; driven by safety concerns and regulatory pressures to limit gun rights
HC Access	-0.13	Moderate decrease; treatment demand, bed saturation, and emergency utilization strains capacity
HC Cost	+0.32	Heavy increase; ER visits, inpatient care, chronic disease, and long-term treatment burden
HC Quality	-0.15	Moderate decline; overload and case complexity reduce capacity and care effectiveness
Homelessness	+0.18	Moderate increase; financial instability and social breakdown raise the risk of housing loss
Immigration	-0.04	Negligible effect; slightly constrains admissibility and retention through policy filters
Innovation	-0.16	Moderate suppression: innovation; impaired focus, consistency, and sustained problem-solving
Law Enforcement	+0.20	Substantial increase; it raises LE demand through overdose response, disorder, and drug-related crime
Liberty	-0.22	Substantial decrease; compulsive behavior erodes autonomy and self-directed choice
Market Freedom	-0.11	Slight decrease; impairs agency and exploitative incentives distort voluntary exchange
Mental Health	-0.35	Heavy decrease; reinforces depression, anxiety, and stress disorders through comorbid feedback loops
Military	-0.16	Moderately reduces recruitment eligibility, readiness, and operational performance
Poverty	+0.24	Substantial increase; reduces employability and degrades financial responsibility
Public Trust	-0.12	Slight decrease; early stigma and perceived unpredictability reduce interpersonal trust
Rule of Law	-0.22	Substantial decrease; inconsistency, ineffectiveness, and perceived illegitimacy reduces integrity
Sickness/Disease	+0.32	Heavy increase; direct physiological harm, comorbidities, and general health burden
Spirituality/God	-0.17	Lorem ipsum dolor sit , consectetur adipiscing elit. Aenean lorem felis, pharetra at sapien at, vulputate
Welfare	+0.22	Lorem ipsum sit amet, consectetur adipiscing elit. Aenean lorem felis, pharetra at sapien at, vulputate fringilla .

2 Agriculture

AGRICULTURE is much more than the total seasonal harvest projected into a commodity chart that a hedge-fund manager can present in a boardroom or an economic conference; it is the quiet mother of civilization itself. Rather than a simple capacity, it is a dense system variable: fields and livestock, management and labor, machines and ecological sustainability, cross-sector coordination, soil, logistics, knowledge, capital dynamics, technological adoption, and the living choreography that keeps agricultural products flowing through store shelves and onto the tables of hundreds of millions of families around the world.

In other words, it asks not only how much the system produces, but how well it works, how stable it is, and whether it can keep working under pressure.

» AGRICULTURE includes:

- › Productive capacity of crops and livestock
- › Quality of labor within agricultural systems
 - Agricultural workforce availability
 - Health, skill level, and productivity
- › Mechanization, tools, and technological integration
- › Infrastructure and logistical reliability
- › Environmental Sustainability and soil vitality
- › Community-level agricultural cohesion and knowledge transmission

» AGRICULTURE exludes:

- › Broader industrial output
- › Climate dynamics
- › National wealth outside agricultural systems
- › Agricultural regulations, laws, or policies

Agriculture is a foundational stabilizer in the model because it is a foundational stabilizer in reality. When agricultural systems are strong, food security, economic stability, rural communities, and public health tend to stand upright with it. When agriculture weakens, fragility spreads outward in quiet but compounding ways.

Research Metrics & Methodology

Primary	Secondary
Crop yield stability	Input efficiency (fertilizer, energy, water, etc.)
Livestock productivity	Income stability
Food affordability pressure	Post-harvest loss rates
Agricultural labor productivity	Soil health and land degradation
Production-storage-distribution continuity	Import dependence and commodity-shock exposure

Data Challenges

- › Subsistence and informal production are often under-counted
- › Cross-country reporting standards are inconsistent
- › Climate shocks can obscure structural agricultural performance
- › Policy and subsidy regimes can distort market-facing indicators
- › Many effects are lagged, so short windows can misrepresent direction and strength

These limitations mean Agriculture should be interpreted as a systems variable, not a single-output statistic. Short-term output can improve while structural resilience weakens, or vice versa.

Known Confounds & Academic Disputes

- › High-output industrial models vs long-run ecological sustainability
- › Scale efficiency vs resilience of distributed/local food systems
- › Direction of causality in development loops (Poverty:low → Agriculture vs. Agriculture → prosperity:low)
- › Institutional quality and infrastructure acting as hidden mediators in social outcome pathways

Because research does not always agree, the influence values in Society V1 are not copied from any single study. They are informed by broader mechanisms, consistent directional signals, and the clearest patterns that appear across multiple sources. As better evidence emerges, the values can be adjusted and refined.

2.1 Birthrate

description here

2.2 Corruption

description here

2.3 Cost of Living

description here

2.4 Crime

description here

2.5 Death Rate

description here

2.6 Discrimination

description here

2.7 Diversity

description here

2.8 Education Access

description here

2.9 Education Cost

a

2.10 Education Quality

a

2.11 Employment

a

2.12 Energy Production

a

2.13 Equality

a

2.14 Family Orientation

a

2.15 Freedom of Speech

a

2.16 Gun Rights

a

2.17 Healthcare Access

a

2.18 Healthcare Cost

a

2.19 Healthcare Quality

a

2.20 Homelessness

a

2.21 Immigration

includes legal retention / admissibility / long-term settlement

- 2.22 Infrastructure
 - a
- 2.23 Innovation
 - a
- 2.24 Law Enforcement
 - a
- 2.25 Liberty
 - a
- 2.26 Market Freedom
 - a
- 2.27 Mental Health
 - a
- 2.28 Military
 - a
- 2.29 Minimum Wage
 - a
- 2.30 Pollution
 - a
- 2.31 Poverty
 - a
- 2.32 Public Trust
 - a
- 2.33 Rule of Law
 - a
- 2.34 Sickness/Disease
 - a
- 2.35 Spirituality/God
 - a
- 2.36 Taxes
 - a
- 2.37 Technology
 - a
- 2.38 Transportation
 - a
- 2.39 Welfare Support
 - a



3 Template Sheet (replace with variable name)

NAME's description

» NAME includes:

- » 1
- » 2
- » 3
- » 4
- » ...

» What this variable excludes:

- » 1
- » 2
- » 3
- » 4
- » ...

Model description for NAME

Research Metrics & Methodology

Primary	Secondary
p1	s1
p2	s2
p3	s3
p4	s4
p5	s5

Data Challenges:

- 1
- 2
- 3
- 4
- 5
- 6

Known Confounds & Literature Disputes

- 1
- 2
- 3
- 4
- 5